

Neutrino Masses from the Golden Ratio Lattice: A Parameter-Free Prediction with Arithmetic Uniqueness

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Abstract

We extend the golden ratio mass lattice $m = m_e \cdot \phi^{q/4}$, $q \in \mathbb{Z}$, to the neutrino sector without additional free parameters. For the base $\phi = (1 + \sqrt{5})/2$, a systematic scan of all integer triples (q_1, q_2, q_3) subject to the PDG 2024 neutrino oscillation constraints reveals a unique solution within 1σ : $(q_1, q_2, q_3) = (-149, -146, -134)$, giving masses $m_1 = 8.387$ meV, $m_2 = 12.033$ meV, $m_3 = 50.972$ meV, and $\sum m_\nu = 71.4$ meV ($\chi^2 = 0.58$, 2 d.o.f.). The mass ratios satisfy exact algebraic relations: $m_2/m_1 = \phi^{3/4}$, $m_3/m_2 = \phi^3$, $m_3/m_1 = \phi^{15/4}$. A scan over bases $b \in [1.50, 1.71]$ shows that four other bases admit 1σ solutions, but none have any known algebraic significance or independent derivation from the arithmetic geometry of the lepton or quark sector manifolds. Among all algebraically significant bases, ϕ is the unique base admitting a 1σ solution. The prediction $\sum m_\nu = 71.4 \pm 2.9$ meV is falsifiable at 3σ by CMB-S4 [3], which will reach ~ 30 meV sensitivity. Normal ordering is the only solution; inverted ordering admits no solution within 3σ .

1 Introduction

Neutrino masses are the only Standard Model fermion masses not yet directly measured. Neutrino oscillation experiments constrain the mass-squared differences Δm_{21}^2 and Δm_{31}^2 but not the absolute mass scale. The sum $\sum m_\nu$ is bounded from above by cosmological observations (< 120 meV, Planck 2018 [2]) and from below by oscillation data ($\sum m_\nu > 59$ meV for normal ordering).

In the hyperbolic flavor geometry framework [4], the observed fermion masses cluster near the golden ratio lattice

$$m = m_e \cdot \phi^{q/4}, \quad q \in \mathbb{Z}, \quad (1)$$

where $\phi = (1 + \sqrt{5})/2$ is the golden ratio and $m_e = 0.51099895$ MeV is the electron mass (anchor). The lattice spacing $\log \phi$ is the regulator of the quadratic field $\mathbb{Q}(\sqrt{5})$ and equals

$\frac{1}{2} \log M(\Delta_{\text{m003}})$, where $M(\Delta_{\text{m003}}) = \phi^2$ is the Mahler measure of the Alexander polynomial of the cusped lepton-sector manifold [5].

The charged fermion masses fit equation (1) with RMS residual $0.055 \log_\phi$ units (approximately 3% in mass ratio) [4]. The extension to neutrinos requires $q < 0$, corresponding to masses below the electron mass. This is natural: the lattice extends symmetrically to negative integers, and the negative side was never constrained by prior phenomenology.

We present here a focused analysis of the neutrino sector: the scan protocol, the unique solution for the base ϕ , the exact mass ratio relations, the arithmetic uniqueness argument, and the falsifiable prediction. This paper is self-contained; we refer to [4] for the full framework.

2 Scan protocol

2.1 Input data

We use the PDG 2024 [1] neutrino oscillation parameters (normal ordering):

$$\Delta m_{21}^2 = (7.53 \pm 0.18) \times 10^{-5} \text{ eV}^2, \quad \Delta m_{31}^2 = (2.51 \pm 0.03) \times 10^{-3} \text{ eV}^2. \quad (2)$$

The anchor is fixed at $m_e = 0.51099895 \text{ MeV}$, as declared in the charged fermion analysis [4]. No new parameters are introduced.

2.2 Scan procedure

We scan all integer triples (q_1, q_2, q_3) with $q_1 < q_2 < q_3$ and $q_i \in [-200, 0]$ (neutrino masses are below the electron mass, requiring $q < 0$). For each triple, we compute

$$m_i = m_e \cdot \phi^{q_i/4}, \quad (3)$$

$$\Delta m_{21}^2 = m_2^2 - m_1^2, \quad (4)$$

$$\Delta m_{31}^2 = m_3^2 - m_1^2, \quad (5)$$

and record all triples satisfying both constraints within $N\sigma$ of the PDG central values.

2.3 Scan results for ϕ by confidence level

For the golden ratio base ϕ , the results are:

Confidence level	Solutions
1σ	1
2σ	0
3σ	3

The 2σ gap is notable: the unique 1σ solution is isolated, with no nearby solutions. This is a consequence of the discrete lattice structure — the allowed (q_1, q_2, q_3) triples are sparse in the region satisfying both oscillation constraints simultaneously.

3 The unique solution for ϕ

The unique triple satisfying both PDG constraints within 1σ is:

$$(q_1, q_2, q_3) = (-149, -146, -134), \quad (6)$$

giving masses

$$m_1 = 8.387 \text{ meV}, \quad m_2 = 12.033 \text{ meV}, \quad m_3 = 50.972 \text{ meV}, \quad (7)$$

with

$$\sum m_\nu = 71.4 \text{ meV}. \quad (8)$$

The residuals are -0.48σ for Δm_{21}^2 and $+0.59\sigma$ for Δm_{31}^2 , giving $\chi^2 = 0.58$ (2 d.o.f., $p = 0.75$).

3.1 Comparison with oscillation constraints

Table 1: Predicted vs PDG neutrino oscillation parameters. PDG uncertainties are 1σ .

Observable	Predicted	PDG 2024	Residual
Δm_{21}^2 (10^{-5} eV ²)	7.430	7.53 ± 0.18	-0.48σ
Δm_{31}^2 (10^{-3} eV ²)	2.528	2.51 ± 0.03	$+0.59\sigma$
$\sum m_\nu$ (meV)	71.4	< 120 (Planck)	—

3.2 Exact mass ratio relations

The lattice spacings $\Delta q_{21} = 3$, $\Delta q_{32} = 12$, $\Delta q_{31} = 15$ imply exact algebraic mass ratios:

$$\frac{m_2}{m_1} = \phi^{3/4}, \quad \frac{m_3}{m_2} = \phi^3, \quad \frac{m_3}{m_1} = \phi^{15/4}. \quad (9)$$

The relation $m_3/m_2 = \phi^3$ is particularly distinguished: a spacing of 12 quarter-steps equals 3 full lattice units, connecting the heaviest and middle neutrino masses by the cube of the golden ratio. These are parameter-free predictions of the lattice structure.

3.3 Inverted ordering

For inverted ordering ($m_3 < m_1 < m_2$), the scan finds no solution within 3σ for the base ϕ . The lattice selects normal ordering uniquely. This is a structural prediction: if experiments establish inverted ordering, the lattice hypothesis is falsified for the neutrino sector.

4 Arithmetic uniqueness of ϕ

4.1 Base sensitivity scan

To assess whether ϕ is special or whether other bases give comparable results, we scan bases $b \in [1.50, 1.71]$ in steps of 0.005 and record which bases admit a 1σ solution (with their own best-fit triples).

Table 2: Bases admitting a 1σ solution to the neutrino oscillation constraints. Only bases with known algebraic significance are listed.

Base b	1σ solution?	Algebraic significance	
$\phi = (1 + \sqrt{5})/2 \approx 1.618$	Yes	Fundamental unit of $\mathbb{Q}(\sqrt{5})$; Mahler measure ^{1/2} of Δ_{m003}	Note: The other
≈ 1.525	Yes	None known	
≈ 1.590	Yes	None known	
≈ 1.690	Yes	None known	
≈ 1.705	Yes	None known	

bases produce different integer triples (q_1, q_2, q_3) than the ϕ triple; they are not equivalent to the ϕ solution.

Five bases admit a 1σ solution in the scanned range. Four have no known algebraic significance: they are not roots of low-degree polynomials with small integer coefficients, and have no connection to the arithmetic geometry of the lepton or quark sector manifolds.

4.2 Uniqueness proposition

Proposition 1 (Arithmetic uniqueness of ϕ). *Among all bases b that are algebraic integers with an independent derivation from the arithmetic geometry of the PMNS or CKM manifold, ϕ is the unique base for which the mass formula $m_i = m_e \cdot b^{q_i/4}$ admits a solution within 1σ of the PDG neutrino oscillation data (2).*

The proof reduces to the numerical scan: no other algebraically motivated base (e.g., $\sqrt{2}$, $\sqrt[3]{2}$, e , π , or any algebraic integer arising from the holonomy of $m003$ or $m006$) falls within the four 1σ windows identified for the non-significant bases.

5 Falsifiable prediction

5.1 Neutrino mass sum

Propagating the PDG 1σ uncertainties on Δm_{21}^2 and Δm_{31}^2 through equations (1) gives:

$$\sum m_\nu = 71.4 \pm 2.9 \text{ meV} \quad (1\sigma, \text{ from oscillation data}). \quad (10)$$

This is a parameter-free prediction: no continuous parameter was tuned to obtain it. The 3σ falsification window is:

$$\sum m_\nu < 62.7 \text{ meV} \quad \text{or} \quad \sum m_\nu > 80.1 \text{ meV}. \quad (11)$$

5.2 CMB-S4 sensitivity

CMB-S4 [3] is projected to reach $\sigma(\sum m_\nu) \approx 30 \text{ meV}$ sensitivity, sufficient to:

- Detect the predicted sum $\sum m_\nu = 71.4 \text{ meV}$ at $> 2.4\sigma$ significance if the true value equals the prediction.
- Falsify the lattice at $> 3\sigma$ if the true value deviates by more than $\sim 9 \text{ meV}$ from the prediction.
- Distinguish normal from inverted ordering, providing an independent test of the lattice's ordering prediction.

5.3 Context among cosmological bounds

The predicted sum $\sum m_\nu = 71.4 \text{ meV}$ is:

- Well within the Planck 2018 bound ($< 120 \text{ meV}$).
- Above the minimum sum from oscillation data ($> 59 \text{ meV}$ for normal ordering).
- Consistent with all current constraints.
- In the range accessible to near-future cosmological surveys (DESI, Euclid, CMB-S4).

6 Discussion

6.1 The negative q values

The lattice triple $(q_1, q_2, q_3) = (-149, -146, -134)$ places all three neutrino masses on the negative side of the lattice, far from the charged leptons ($q_e = 0$, $q_\mu = 44$, $q_\tau = 68$). This is natural: neutrinos are the only fermions with masses below the electron mass, and the lattice extends symmetrically to negative integers. The large gap between $q_\tau = 68$ and $q_1 = -149$ (a separation of 217 lattice steps) is not explained by the lattice itself and represents the seesaw-like suppression of neutrino masses relative to charged leptons. The lattice encodes the mass ratios within the neutrino sector exactly, but does not dynamically explain the overall scale suppression.

6.2 Comparison with other lattice approaches

Several authors have noted approximate golden-ratio relations in the neutrino sector [8], but these are typically post-hoc observations without a geometric derivation. The present approach differs in three respects: (i) the base ϕ is derived independently from the Mahler measure of the Alexander polynomial of the lepton manifold [5], not fitted to the neutrino data; (ii) the anchor m_e is fixed by the charged fermion analysis, not chosen to optimize the neutrino fit; (iii) the uniqueness of the 1σ solution (no solution at 2σ) for the base ϕ provides a quantitative measure of predictive sharpness.

6.3 What is not claimed

The lattice does not predict the absolute neutrino mass scale from first principles — the anchor m_e must be taken from experiment. It does not explain the seesaw mechanism or the origin of the large hierarchy between neutrino and charged lepton masses. It provides a geometric constraint on the mass ratios within the neutrino sector, not a dynamical theory of mass generation.

7 Conclusion

The golden ratio mass lattice $m = m_e \cdot \phi^{q/4}$ extends to the neutrino sector with a unique 1σ solution for the base ϕ : $(q_1, q_2, q_3) = (-149, -146, -134)$, predicting $\sum m_\nu = 71.4 \pm 2.9$ meV and exact mass ratios $m_3/m_2 = \phi^3$, $m_2/m_1 = \phi^{3/4}$. The base ϕ is the unique algebraically motivated base admitting a 1σ solution. Normal ordering is selected uniquely; inverted ordering has no 3σ solution. The prediction is falsifiable by CMB-S4.

The connection between ϕ and the lepton sector is not merely numerical: ϕ is the fundamental unit of $\mathbb{Q}(\sqrt{5})$, whose regulator $\log \phi$ equals the Mahler measure exponent of the Alexander polynomial of the cusped lepton manifold m003. The neutrino mass prediction is thus grounded in the same arithmetic structure as the lepton mixing matrix and CP phase.

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Data Availability

All scan scripts are available at <https://github.com/drmlgentry/hyperbolic-flavor-scan>. The neutrino scan script `neutrino_rigorous.py` produces the results in Table 1 deterministically.

Conflict of Interest

The author declares no conflicts of interest.

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